CLUSTERING using kmeans ++

* Code

- Agenda ⇒ Recap, Dunn index, k, problem, scratch imp., kmeans ++

Lloyds algo

Recap → (1) initialize centroids Randomly

k = # centroids

(2) for i in range(max_iter):

100
200

(2) assign DP to nearest centroid

(3) update centroids → avg

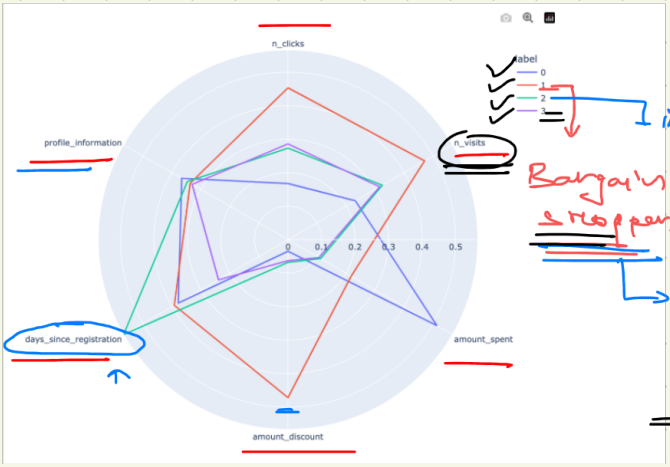
tol → tolerance = 0.001 <=

stop

* > → continue updating centroid

SS

(4) POST CLUSTER Analysis



data

⇒ plotly

inactive old users

Bargain shoppers

polar plot

Show discounted items

⇒ polar plot

k=4 k=3 k=5

Real

code

Mathematical formulation ⇒

obj func → clusters to have similar pp

k clusters → k centroids

pp

given $D = \{x_i\}_{i=1}^{N}$

To find
- k-clusters → S_1 S_2 S_3 ... S_k
- k centroids → c_1 c_2 ... c_k

OR J func^n

K → no. of clusters

$$\underset{c_1 c_2 c_3 \cdots c_K}{\operatorname{argmin}} \sum_{j=1}^{K} \sum_{x_i \in S_j} \|x_i - c_j\|^2$$

intra cluster dist^n

c_j → centroids

Cluster loc^n → 0, 1, 2, 3

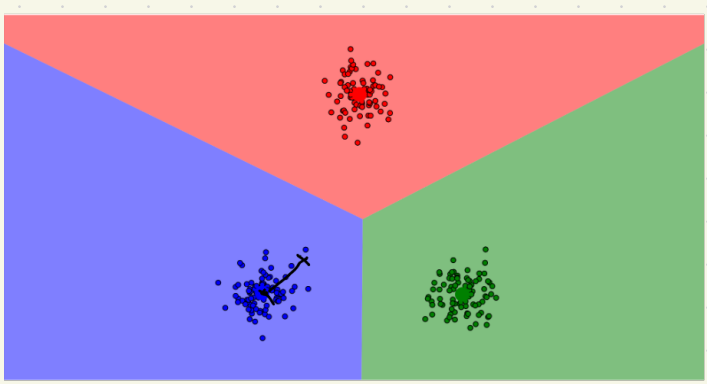
centroid

$j=1$ → Cluster 1 → dist^n

$j=2$ → another c → CLUSTER

intra dikhi

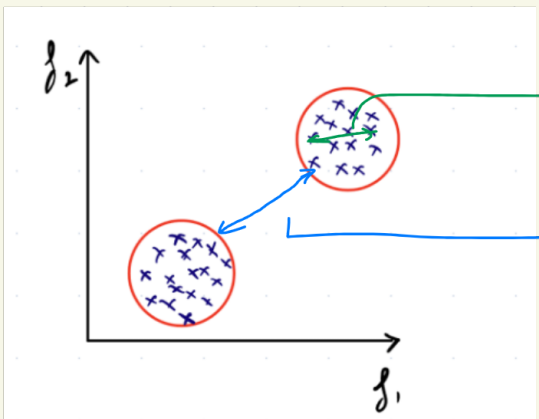
$(\text{dist})^2$



within cluster dists → large or small

Small

* Evaluating a CLUSTER
  - Business
  - Viz → TSNE → plot



INTRA CLUSTER

INTER CLUSTER distance

metric = inter cluster ↑
intra cluster ↓

(1) DUNN index →

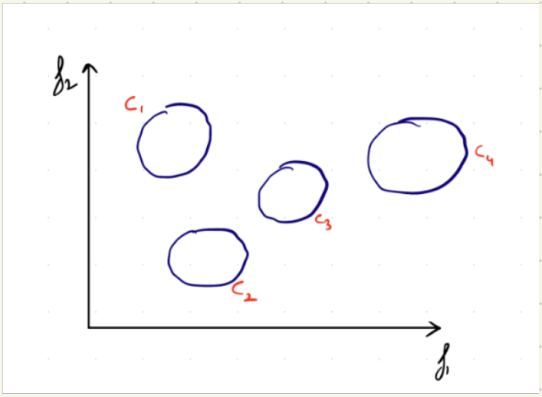
min

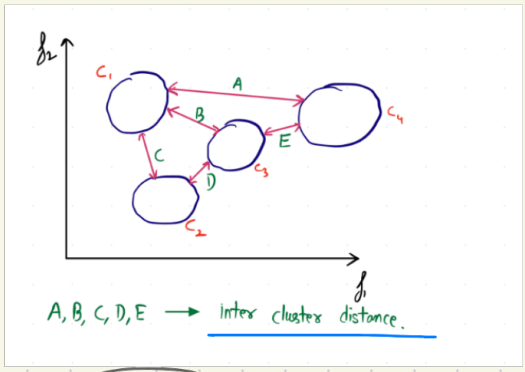
(2) $Dunn = \frac{\min_{i,j} dist(i,j)}{\max_k dist(k)}$

Higher Dunn

DUNN

inter



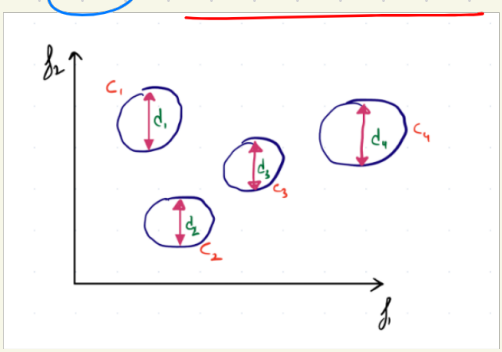


A, B, C, D, E → inter cluster distance.

Dunn

SHORTEST

ex intra cluster



largest ~~any~~

Dunn index

C_i & C_j

ex $D = \dfrac{\min_{i,j} dist(i,j)}{\max_k dist(k)}$

how bad

Dunn index → Var.

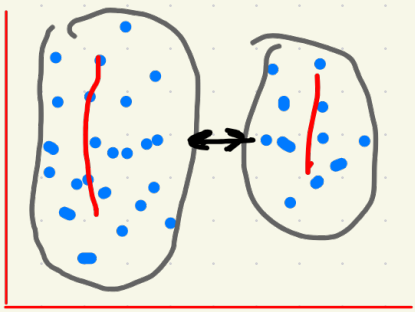
↑ ⟹ ⌄

$$D = \frac{\min_{i,j} distance(i, j)}{\max_k distance'(k)}$$

shortest inter cluster dist

largest intra cluster

k

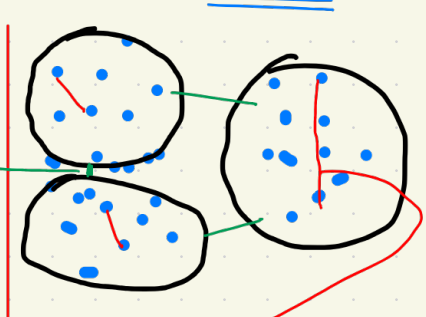


④ DUNN index

3 CLUSTERS

worst ⇒

min



dunn index

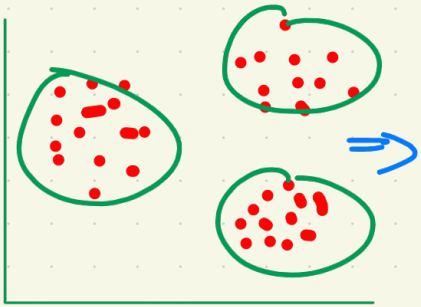
max.

k=3 → DUN

k=4 → DUN

k=5 → DUN

Higher Dunn ?





High

Dunn ↑

Dum ↓ ⇒

Evaluating overall quality

0/1 clustering

Range of Dunn index

$$D = \frac{\min_{i,j} distance(i,j)}{\max_k distance'(k)}$$

max = ∞

min = 0

min = 0

DUNN

max → ∞

min → 0

0, ∞ ⇒

3 min

10

select Best value of K

Visualize

Viz^n → X

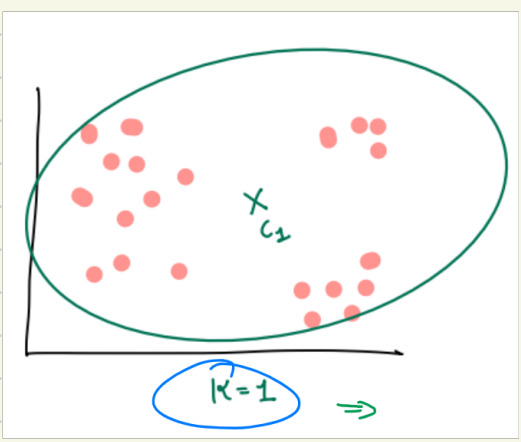
PCA, TSNE → Viz^n

2 Business / Domain → K=4

K=5

3 Elbow method

no. of k



→ good ?

X

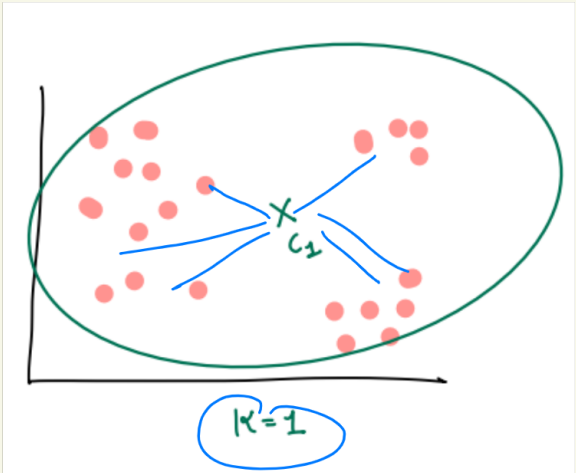
justify

→ intra cluster dist

DUNN →

WCSS = intra cluster dist

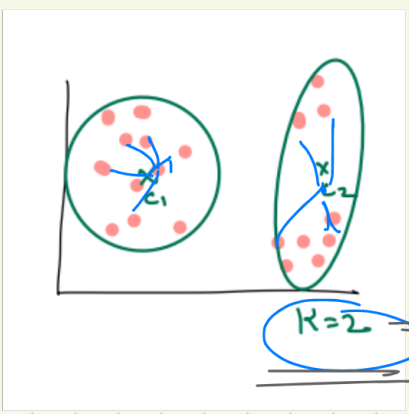
↳ within cluster sum of sq. of distance



sq

dist

WCSS ↑↑

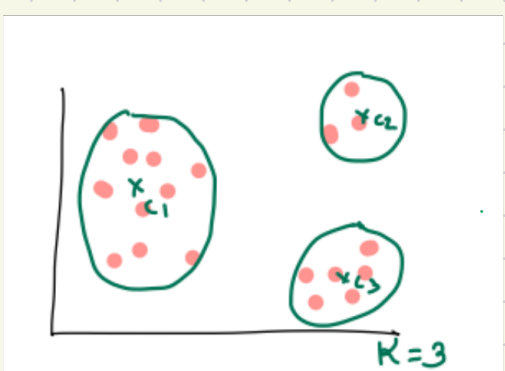


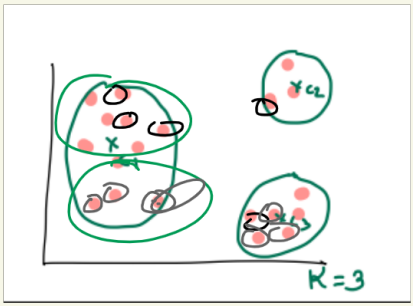
WCSS = ?

WCSS(K=1) V/s

WCSS(K=2)

WCSS(K=3) ↓↓





k = 4

↑↑ wcss ↓↓

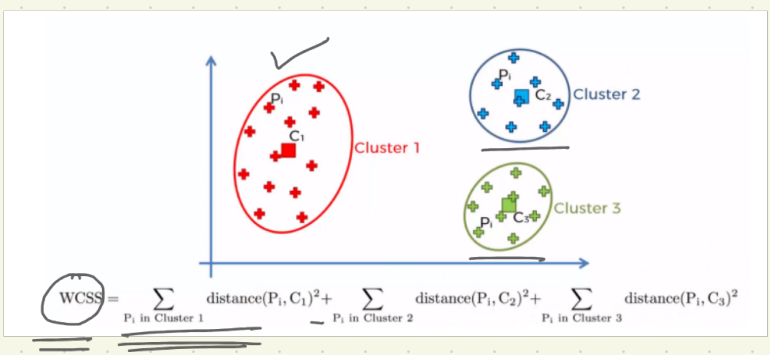
continue till k = n

wcss → inertia

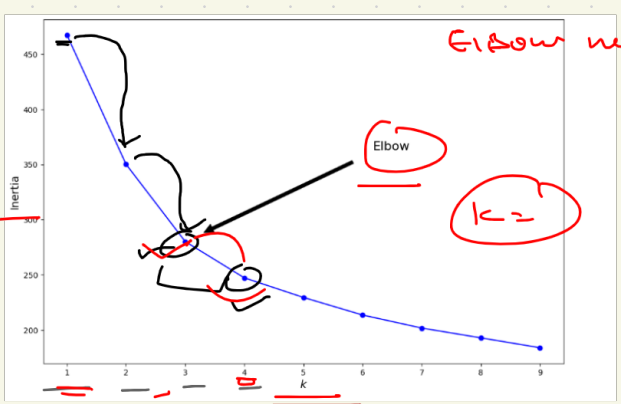
= 0

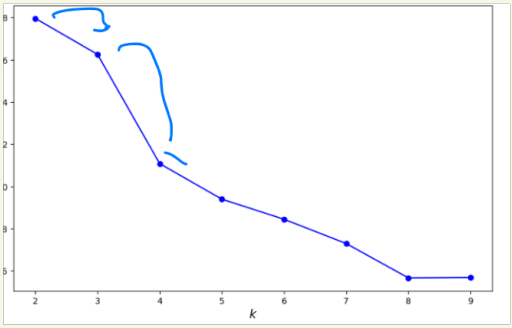
wcss = 0 ✗✗

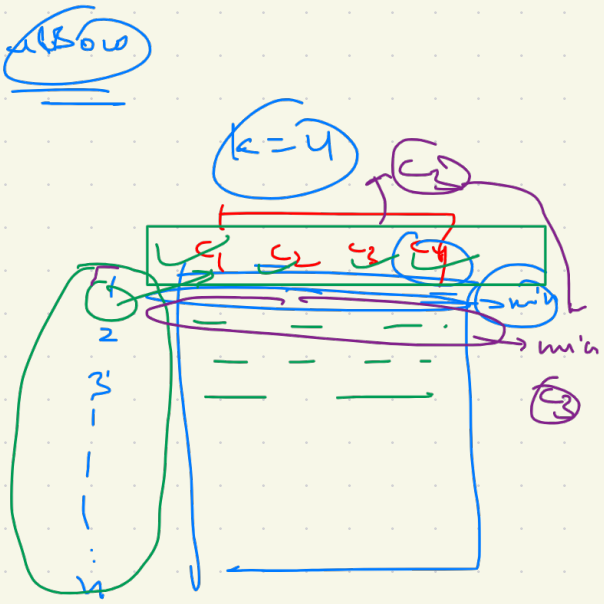
k = n



wcss → inertia







(8) Time complexity of kmeans

(n), (d), (k), (i)

x_i = [Age, income]

(1) Assign

(2) Recalculation of centroids

Step 1 =>













$$\sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2 + (x_d - c_d)^2}$$

(dist $\rightarrow$ $O(d)$)

$O(nkd)$ Step 1

Step 2 Cluster 1 $\rightarrow$ (x_1, y_1) , - - -

avg

new centroid = (k pp) d from

$O(nd) + O(kd)$

$= O(nd)$

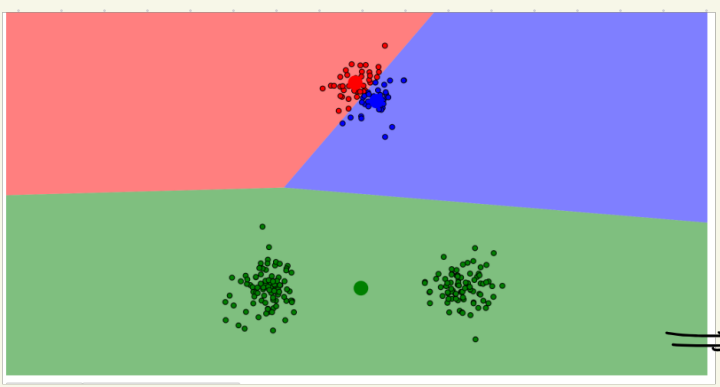
1 iteration →

$O(nkd) + O(nd)$

$= O(nkdi)$

$O(nkdi)$

$O(nd)$

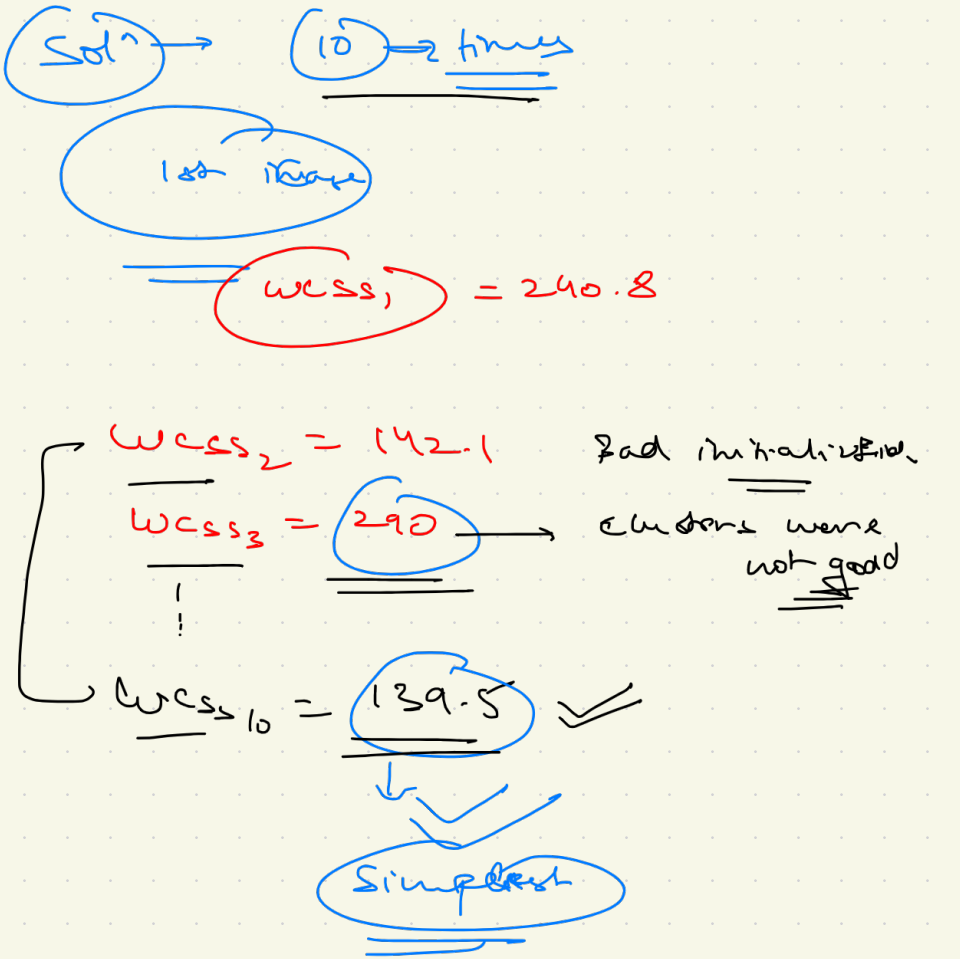


→ Not optimal clustering

Random

Centroid → smarter way

Initialization Trap



Next / consequent centroid will be farthest points from the initial centroid

final soln

kmeans ++

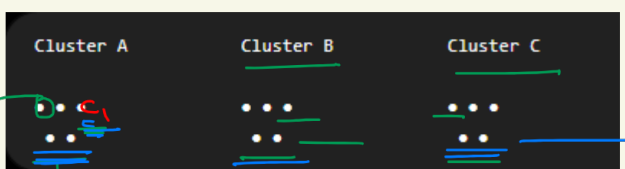
(1) 1st centroid @ Randomly (C_1)

(2) compute dist from C_1 to Rest of points

(3) Pick (C_2) such that

prob. of C_2 ∝ distance C_1

repeat = k



very distant

Small dist.

large dist.

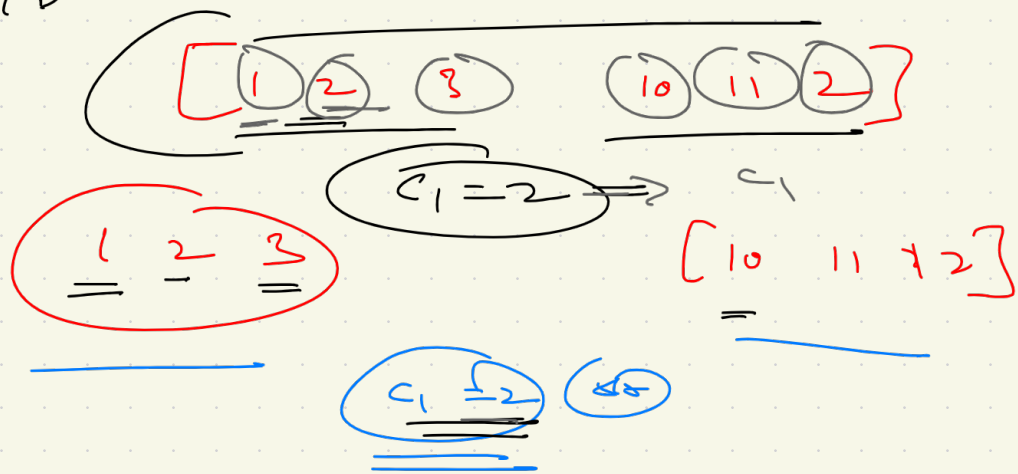
C_2

Step 3

$$P(x) \propto \frac{D(x)^2}{\sum_i D(x)^2}$$



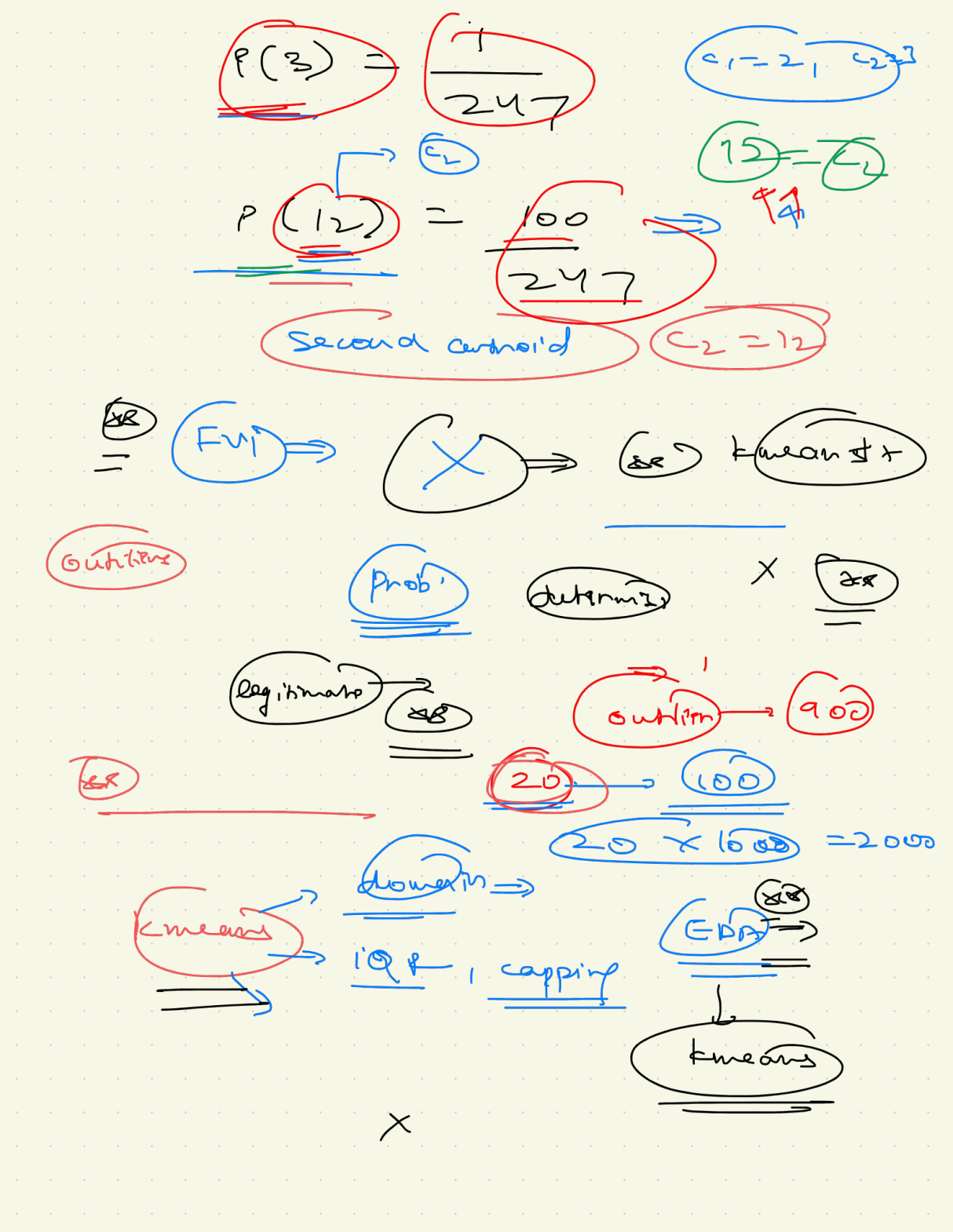
1D



Dist from c_1

Point	Dist.	Sq. Dist
1	1	1
2	0	0
3	1	1
10	8	64
11	9	81
12	10	100

Sum = 247

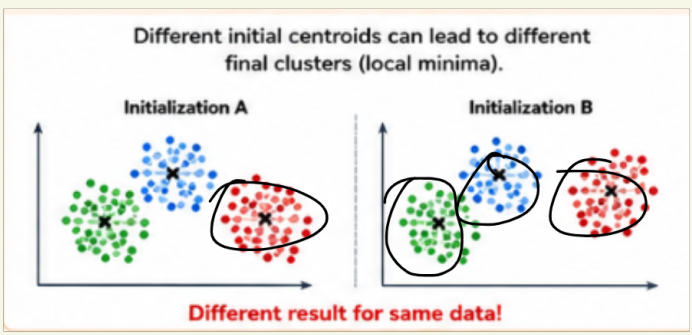

P(3) = 1/247
c1 = 2, c2 = 3
12 = c2
P(12) = 100/247
Second centroid
c2 = 12
Outliers
Prob'
deterministic
Regression
outlier
900
20
100
20 x 100 = 2000
Kmeans
clustering
IQR, capping
EDA
Kmeans

Limitations of kmeans ⇒

next

1. Specify k

2. Sensitive to initialization



⇒

Feature Scaling

Globular shapes

Outliers

Categorical

GMMs

Hierar

Doubts